ACT SUBJECT TEST & DIGITAL SAT MATH COURSES

Rabaa Farrag

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ACT2 • ACT SUBJECT TEST MATH LEVEL 1 • PRACTICE EXAM

ACT2 Exam 1

Full Practice Test • Test #1

Algebra, geometry, functions, probability & spatial reasoning. Work through all 50 questions under timed conditions, then check your answers with the answer key and the step-by-step solutions at the end.

50	MCQ	60 min
QUESTIONS	MULTIPLE CHOICE	TIME LIMIT

NAME _____ DATE _____ SCORE _____ /50

How to get the most from this practice test

- **Read** every question slowly and underline the key words.
- **Draw** a quick sketch for every geometry question before you solve it.
- **Solve** step by step and show your work.
- **Check** your answer by substituting it back before you choose.

RF

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Eng. Rabaa Farrag

ACT, Digital SAT & EST Math expert with **23 years of experience**, helping students master math and reach **700+**.

ACT SUBJECT TEST • MATH LEVEL 1 • TEST #1

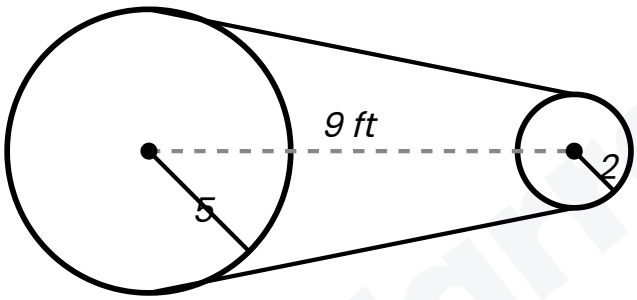
Practice Questions

50 QUESTIONS

1 The perimeter of a square is equal to the circumference of a circle whose area is $20\ \text{in}^2$. What is the area in square inches of the square?

(A) 5π
(B) 5
(C) 20π
(D) 20

2 The large circle has a radius of 5 ft and the small circle has a radius of 2 ft. The distance between the two centers of the circles is 9 ft. If the large circle rotates 10 times, the small circle rotates how many times?



(A) 4
(B) 10
(C) 25
(D) 50

3 A worker wants to make 10 pounds of mixed nuts using peanuts and cashews. He buys peanuts at 3.75 dollars per pound and cashews at 8 dollars per pound. If 1 pound of his mixture is valued at 5.45 dollars, how many pounds of peanuts did he buy?

(A) 6
(B) 10
(C) 7.25
(D) 8

4 A garden with measurements $4\ \text{m} \times 6\ \text{m}$ is surrounded by a sidewalk on all sides. The area of the garden with the sidewalk is $40\ \text{m}^2$, what is the width of the sidewalk?

(A) $\frac{-5+\sqrt{41}}{2}$
(B) $\frac{-5-\sqrt{41}}{2}$
(C) 5
(D) 41

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5 Sarah invested 5000 dollars with 3% simple annual interest rate. What will be the interest in Sarah's account after t years?

(A) $15t$
(B) $150t$
(C) $3t$
(D) $1.5t$

6 The table below shows the number of bacteria increasing exponentially. What will the value of y be when $x = 10$?

X	Y
0	1
1	2
2	4
3	8

(A) 124
(B) 1024
(C) 2015
(D) 200

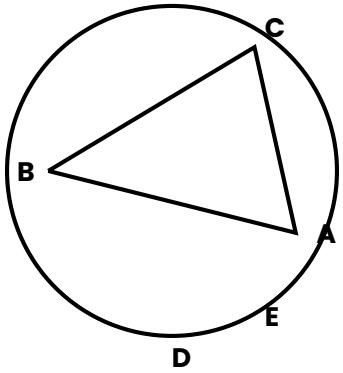
7 If $X = 2Y$, which of the following is the value of X?



(A) 60
(B) 120
(C) 30
(D) 45
(E) 100

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8 If Measure of Arc $AED = 84°$, what is the Measure of Angle ACE?



(A) 24
(B) 84
(C) 42
(D) 168

9 Jade bought two similar planters. The second planter has dimensions 3 times as large as the first planter. If the first planter gets filled using 1 bag of soil, how many bags will the second planter need to be filled?

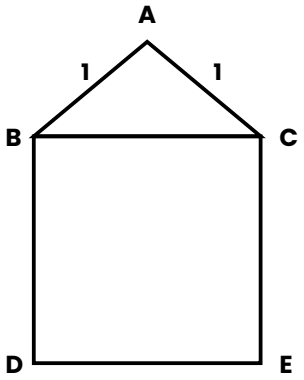
(A) 9
(B) 81
(C) 27
(D) 3

10 If $f(x) = y$, a point with coordinates (x, y) lies if the first number of the coordinates lies 3 less than half the second number?

(A) $y = \frac{1}{2}x - 3$
(B) $y = 2x - 3$
(C) $y = \frac{1}{2}x + 3$
(D) $y = 2x - 3$

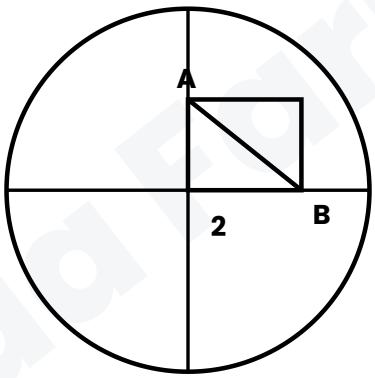
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11 Triangle ABC is isosceles. Find the perimeter of $ABCED$.



(A) $2+\sqrt{3}$
(B) $2+3\sqrt{2}$
(C) $2\sqrt{3}$
(D) $3\sqrt{2}$

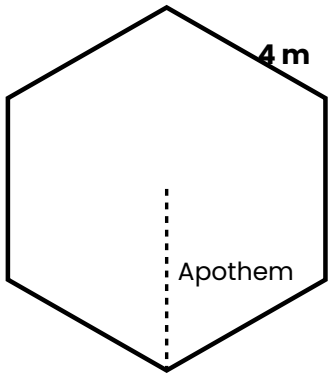
12 What proves $AB=2$?



(A) Diagonal of the rectangle are congruent
(B) The radius $=2$
(C) The second diagonal $=$ the radius
(D) All the above

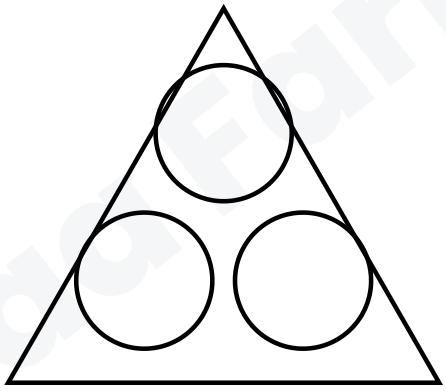
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13 Find the area of a regular hexagon with side length 4 m. Round to the nearest tenth.



(A) $83.1\ \text{m}^2$
(B) $24\ \text{m}^2$
(C) $41.6\ \text{m}^2$
(D) $20.8\ \text{m}^2$

14 In this figure each of the 3 circles is tangent to the other. Each side of the triangle is tangent to 2 of the circles. Given that each circle has a radius of 3, what is the perimeter of the triangle?



(A) $18 + 18\sqrt{2}$
(B) $18 + 18\sqrt{3}$
(C) $36 + 18\sqrt{2}$
(D) $36 + 36\sqrt{2}$

15 Ali invested \$5000 in an account that paid him 32% compound interest, what will the balance of his account be after t years?

(A) $1600t$
(B) $5000 + 1600t$
(C) $5000(132)t$
(D) $5000(1.32)^t$

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16 What is the value of y for the solution to this system of equations $\begin{cases} 2x - y = 12 \\ y = x + 4 \end{cases}$?

A 8
B 16
C 20
D 24

17 Simplify the following expression $-6x - (8 \div 2) - (4x - 5)$

A $-10x + 1$
B $-10x + 9$
C $-2x - 9$
D $-2x + 1$

18 If x and y are integers, which equation is always true?

A $\frac{x}{y} = \frac{y}{x}$
B $x - y = y - x$
C $x + 2y = y + 2x$
D $2x + 3y = 3y + 2x$

19 In the geometric sequence $(16, n, 36, \dots)$, the second term n is the geometric mean of 16 and 36. What is the value of n?

A 24
B 25
C 26
D 28

20 John has x dollars. Ann has \$5 more than John. If Ann gives John \$10, then, in terms of x, how many dollars will Ann have?

A $x - 15$
B $x - 10$
C $x - 5$
D $x + 5$

21 Automobile sales in United States dropped from 10.7 million units one year to 9.3 million the next year. The decrease was closest to what percent?

A 1%
B 13%
C 14%
D 15%

22 If the average of 5 consecutive whole numbers is 12, what is the largest number?

A 14
B 16
C 17
D 22

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23 Solve the following inequality: $5|8-2x|-3>27$

- (A) $x<1$ or $x>7$
- (B) $x\leq 1$ or $x\geq 7$
- (C) $1<x<7$
- (D) $1\leq x\leq 7$

24 Which of the following properties illustrate $3^x\cdot\frac{1}{3^x}=1$?

- (A) Commutative Property of Multiplication
- (B) Associative Property of Multiplication
- (C) Property Multiplicative Inverse
- (D) Identity Property of Multiplication

25 If x is greater than y and y is greater than 1, which of the following is LEAST?

- (A) $\frac{1}{x}$
- (B) $\frac{1}{y}$
- (C) $\frac{1}{x-y}$
- (D) $\frac{1}{y-x}$

26 A store regularly sells books at 20% off the list price. At a sale its regular prices are reduced 10 percent. The sale price is what percent of the list price?

- (A) 30%
- (B) 70%
- (C) 72%
- (D) 79%

27 The price of a cold roast beef lunch is $4.80, and the price of a hot roast beef lunch is $5.40. If both prices include 20 percent for tax and tip, what was the difference in price before tax and tip?

- (A) $0.12
- (B) $0.40
- (C) $0.50
- (D) $0.60

28 For what price is 20 percent off the same as $20 off?

- (A) $1
- (B) $10
- (C) $100
- (D) $1,000

29 To which of the following is 6.06 closest?

- (A) 6.1
- (B) 7
- (C) 6.7
- (D) 6

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30 All of the following are greater than $\frac{1}{2}$ EXCEPT

A $\frac{101}{200}$
B $\frac{17}{33}$
C $\frac{7}{12}$
D $\frac{60}{1000}$

31 If $\frac{1}{4}$ of number is greater than 8, the number must be

A Less than 2
B Equal to 2
C Less than 32
D Greater than 32

32 If Ivan traveled 510 miles in 2 hours, 18 minutes, and 30 seconds. What was Ivan's average speed?

A 221 mile/hour
B 281 mile/hour
C 229 mile/hour
D 310 mile/hour

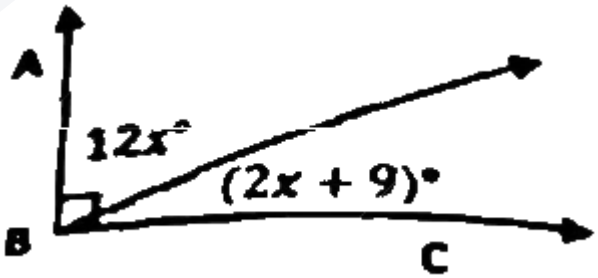
33 Which of the following numbers can be written in the form $(4 \times N) + 1$ where N is a whole number?

A 55
B 62
C 71
D 93

34 If $t = -3$, then $3t^2 + 5t + 6$ equals

A -36
B -6
C 6
D 18

35 Based on the information in the figure, which statement shows a valid argument?



A $(2x + 9) + 12x = 180$ because the angles from a linear pair
B $(2x + 9) + 12x = 90$ because the angles are complementary
C $(2x + 9) = 12x$ because the angles are vertical angles
D $(2x + 9) = 12x$ because the angles are supplementary

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36 According to a market study, City X has 1.75 millionaires for every $1,000$ people. If there are 8 million people in City, how many millionaires are there?

(A) 14
(B) 140
(C) $1,400$
(D) $14,000$

37 What is the sum of the interior angles of a Hexagon?

(A) $720°$
(B) $900°$
(C) $1080°$
(D) $1260°$

38 For the line $x + y = 1$, what is the sum of the values of the slope, x-intercept, and y-intercept?

(A) -1
(B) 0
(C) 1
(D) 2

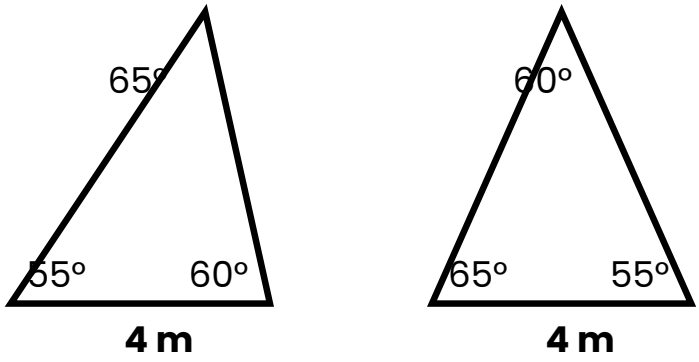
39 Justine and Eric want to know the height of a tree. Eric places a mirror (M) on the ground in line with the top of the tree. He moves towards Justine represented by AB until she can see the top of the tree at point C. Justine's eye is 5 ft from the ground at point A. To the nearest tenth of a foot, What is the height of the tree?



(A) 30
(B) 31
(C) 32
(D) 35

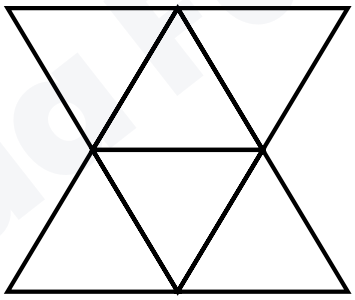
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40 Which statement about these triangles is true?



A The triangles are similar but not congruent
B The triangles are not similar but not congruent
C The triangles are similar and are congruent
D The triangles are neither similar nor congruent

41 The net below represents a regular polyhedron or platonic solid. How many edges does the platonic solid have?



A 6
B 8
C 10
D 12

42 If there are 9 white cards and 7 black cards, what is the probability of randomly picking a white card?

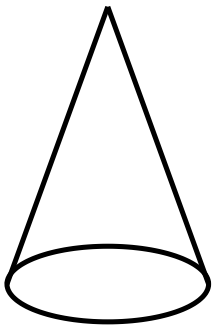
A $\frac{9}{16}$
B $\frac{9}{7}$
C $\frac{7}{9}$
D $\frac{16}{9}$

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43 A circle has a diameter of 11 inches. What is the circumference of that circle?

(A) π
(B) 11π
(C) 5.5π
(D) 121π

44 What is the cross section of this cone?



(A) Circle
(B) Oval
(C) Rectangle
(D) Triangle

45 Simplify $2(2x^3y^5z)^3$

(A) $16x^9y^{15}z^3$
(B) $16x^3y^{15}z^3$
(C) $16x^9y^5z^3$
(D) $16x^9y^{15}z$

46 Simplify $-2pq^3\left(\frac{1}{3}p^2q^5\right)^2$

(A) $-2p^5q^{13}$
(B) $-\frac{2}{9}p^5q^{13}$
(C) $2p^5q^{13}$
(D) $\frac{2}{9}p^5q^{13}$

47 Find the area of a circle with a diameter of 17.

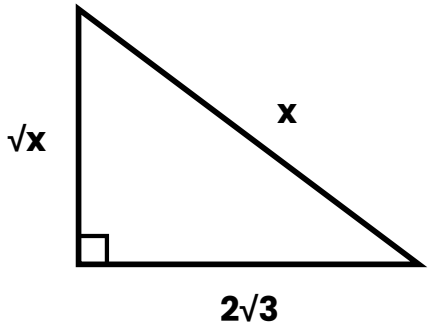
(A) 227
(B) 64
(C) 22.7
(D) 22

48 Factor $25x^2 - 121$

(A) $(5x - 11)(5x - 11)$
(B) $(5x - 11)(5x + 11)$
(C) $(5x + 11)(5x + 11)$
(D) $(x - 11)(x + 11)$

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49 Find the value of x



(A) 2
(B) 3
(C) 4
(D) 6

50 Find the perimeter ratio of the large hexagon to the small one



(A) $\frac{3}{2}$
(B) $\frac{3}{2}$
(C) $\frac{1}{2}$
(D) $\frac{2}{3}$

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TEST #1 • CHECK YOUR WORK

Answer Key

50 ANSWERS

Q	ANSWER	Q	ANSWER
1	A	26	C
2	C	27	C
3	A	28	C
4	A	29	A
5	B	30	D
6	B	31	D
7	E	32	A
8	C	33	D
9	C	34	D
10	A	35	B
11	B	36	D
12	D	37	A
13	C	38	C
14	B	39	D
15	D	40	A
16	C	41	D
17	A	42	A
18	D	43	B
19	A	44	A
20	C	45	A
21	B	46	B
22	A	47	A
23	A	48	B
24	C	49	C
25	D	50	B

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TEST #1 • STEP-BY-STEP

Detailed Solutions

50 SOLUTIONS

1 **Answer: A** 5π

Step 1: Circle area: $\pi r^2 = 20 \Rightarrow r^2 = \frac{20}{\pi} \Rightarrow r = \sqrt{\frac{20}{\pi}}$.
Step 2: Circumference: $C = 2\pi r = 2\pi\sqrt{\frac{20}{\pi}} = 2\sqrt{20\pi}$.
Step 3: The square's perimeter equals C: $4s = 2\sqrt{20\pi} \Rightarrow s = \frac{\sqrt{20\pi}}{2}$.
Step 4: Area of the square: $s^2 = \frac{20\pi}{4} = 5\pi$.

2 **Answer: C** 25

Step 1: The belt makes both circles' rims travel the same distance: (circumference $\times$ turns) is equal for both.
Step 2: Large circle: $2\pi(5)(10) = 100\pi$ ft of belt.
Step 3: Small circle: $2\pi(2)\,n = 100\pi \Rightarrow 4n = 100 \Rightarrow n = 25$ turns. (The 9 ft distance is not needed.)

3 **Answer: A** 6

Step 1: Let p = pounds of peanuts, so cashews $= 10 - p$.
Step 2: Total value of the mixture: $5.45 \times 10 = 54.50$ dollars.
Step 3: $3.75p + 8(10 - p) = 54.5 \Rightarrow 3.75p + 80 - 8p = 54.5$.
Step 4: $-4.25p = -25.5 \Rightarrow p = 6$ pounds of peanuts.

4 **Answer: A** $\frac{-5+\sqrt{41}}{2}$

Step 1: Let w be the sidewalk width. The outer rectangle is $(4 + 2w)$ by $(6 + 2w)$.
Step 2: $(4 + 2w)(6 + 2w) = 40 \Rightarrow 24 + 20w + 4w^2 = 40 \Rightarrow 4w^2 + 20w - 16 = 0 \Rightarrow w^2 + 5w - 4 = 0$.
Step 3: Quadratic formula: $w = \frac{-5\pm\sqrt{25+16}}{2} = \frac{-5\pm\sqrt{41}}{2}$.
Step 4: A width must be positive, so $w = \frac{-5+\sqrt{41}}{2}$.

5 **Answer: B** $150t$

Step 1: Simple interest: $I = Prt$.
Step 2: $I = 5000 \times 0.03 \times t = 150t$ dollars.

6 **Answer: B** 1024

Step 1: From the table, y doubles each time x increases by 1, starting from $y = 1$ at $x = 0$: $y = 2^x$.
Step 2: At $x = 10$: $y = 2^{10} = 1024$.

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7 **Answer: E** 100

Step 1: The three angles of the triangle are $30°$, X and Y, so $30 + X + Y = 180 \Rightarrow X + Y = 150$.
Step 2: Substitute $X = 2Y$: $3Y = 150 \Rightarrow Y = 50°$.
Step 3: $X = 2(50°) = 100°$.

8 **Answer: C** 42

Step 1: $\angle ACE$ is an inscribed angle whose intercepted arc is $\overset{\frown}{AED}$ (arc AE).
Step 2: An inscribed angle equals half its intercepted arc: $\frac{84°}{2} = 42°$.

9 **Answer: C** 27

Step 1: For similar solids, if lengths are multiplied by k, volumes are multiplied by k^3.
Step 2: $k = 3 \Rightarrow$ volume ratio $= 3^3 = 27$.
Step 3: The second planter needs 27 bags of soil.

10 **Answer: A** $y = \frac{1}{2}x - 3$

Step 1: Translate the relationship: the y-coordinate is 3 less than half of the x-coordinate.
Step 2: Half of x is $\frac{1}{2}x$; 3 less than that is $\frac{1}{2}x - 3$.
Step 3: So $y = \frac{1}{2}x - 3$.

11 **Answer: B** $2 + 3\sqrt{2}$

Step 1: $\triangle ABC$ is a right isosceles triangle with legs $AB = AC = 1$.
Step 2: Hypotenuse $BC = \sqrt{1^2 + 1^2} = \sqrt{2}$, which is also the side of square $BCED$.
Step 3: The outer boundary $A \to B \to D \to E \to C \to A$ uses AB, BD, DE, EC, CA: $1 + \sqrt{2} + \sqrt{2} + \sqrt{2} + 1 = 2 + 3\sqrt{2}$.

12 **Answer: D** All the above

Step 1: One diagonal of the rectangle runs from the center to a point on the circle, so it is a radius $= 2$.
Step 2: The diagonals of a rectangle are congruent, so the other diagonal AB also equals 2.
Step 3: All three statements are used, so the answer is "All the above".

13 **Answer: C** $41.6\ \text{m}^2$

Step 1: A regular hexagon is made of 6 equilateral triangles with side $s = 4$.
Step 2: Area of one triangle: $\frac{\sqrt{3}}{4}s^2 = \frac{\sqrt{3}}{4}(16) = 4\sqrt{3}$.
Step 3: Hexagon: $6 \times 4\sqrt{3} = 24\sqrt{3} \approx 41.6\ \text{m}^2$.

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14 **Answer: B** $18 + 18\sqrt{3}$

Step 1: Join the three centers: they form an equilateral triangle of side $2r = 6$, parallel to the outer triangle's sides.
Step 2: Each outer side = 6 (between tangent points) $+2\times$ (corner piece). In each 30-60-90 corner, the piece is $r\sqrt{3} = 3\sqrt{3}$.
Step 3: Side $= 6 + 6\sqrt{3}$, so perimeter $= 3(6 + 6\sqrt{3}) = 18 + 18\sqrt{3}$.

15 **Answer: D** $5000(1.32)^t$

Step 1: Compound interest: $A = P(1 + r)^t$.
Step 2: $P = 5000$, $r = 0.32$: $A = 5000(1.32)^t$.

16 **Answer: C** 20

Step 1: Substitute $y = x + 4$ into $2x - y = 12$: $2x - (x + 4) = 12$.
Step 2: $x - 4 = 12 \Rightarrow x = 16$.
Step 3: $y = 16 + 4 = 20$.

17 **Answer: A** $-10x + 1$

Step 1: Division first: $8 \div 2 = 4$, so $-6x - 4 - (4x - 5)$.
Step 2: Distribute the minus sign: $-6x - 4 - 4x + 5$.
Step 3: Combine like terms: $-10x + 1$.

18 **Answer: D** $2x + 3y = 3y + 2x$

Step 1: Test each choice with numbers, e.g. $x = 1$, $y = 2$: A gives $\frac{1}{2} \neq 2$, B gives $-1 \neq 1$, C gives $5 \neq 4$.
Step 2: D: $2x + 3y = 3y + 2x$ is the commutative property of addition — true for all x, y.

19 **Answer: A** 24

Step 1: In a geometric sequence the middle term is the geometric mean of its neighbours: $n^2 = 16 \times 36 = 576$.
Step 2: $n = \sqrt{576} = 24$ (ratio $\frac{3}{2}$: $16, 24, 36$).

20 **Answer: C** $x - 5$

Step 1: Ann starts with $x + 5$ dollars.
Step 2: After giving John 10 dollars: $(x + 5) - 10 = x - 5$.

21 **Answer: B** 13%

Step 1: Decrease $= 10.7 - 9.3 = 1.4$ million.
Step 2: Percent decrease $= \frac{1.4}{10.7} \times 100\% \approx 13.1\%$, closest to 13%.

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22 **Answer: A** 14

Step 1: For consecutive numbers, the average is the middle number, so the middle one is 12.
Step 2: The numbers are $10, 11, 12, 13, 14$; the largest is 14.

23 **Answer: A** $x < 1$ or $x > 7$

Step 1: $5|8 - 2x| > 30 \Rightarrow |8 - 2x| > 6$.
Step 2: Case 1: $8 - 2x > 6 \Rightarrow -2x > -2 \Rightarrow x < 1$.
Step 3: Case 2: $8 - 2x < -6 \Rightarrow -2x < -14 \Rightarrow x > 7$.
Step 4: Solution: $x < 1$ or $x > 7$.

24 **Answer: C** Property Multiplicative Inverse

Step 1: $\frac{1}{3^x}$ is the reciprocal of 3^x.
Step 2: A number times its reciprocal equals 1: this is the Multiplicative Inverse Property.

25 **Answer: D** $\frac{1}{y-x}$

Step 1: Since $x > y > 1$: $\frac{1}{x}$, $\frac{1}{y}$ and $\frac{1}{x-y}$ are all positive.
Step 2: $y - x < 0$, so $\frac{1}{y-x}$ is negative — the least value.

26 **Answer: C** 72%

Step 1: Regular price $= 80\%$ of list $= 0.80L$.
Step 2: Sale takes another 10% off: $0.90 \times 0.80L = 0.72L$.
Step 3: The sale price is 72% of the list price.

27 **Answer: C** \$0.50

Step 1: Each price includes 20% extra, so price $= 1.20\times$ (base price).
Step 2: Difference of the totals: $5.40 - 4.80 = 0.60$.
Step 3: Difference of the base prices: $\frac{0.60}{1.20} = 0.50$, i.e. \$0.50.

28 **Answer: C** \$100

Step 1: Let the price be P. 20% off saves $0.20P$.
Step 2: $0.20P = 20 \Rightarrow P = \frac{20}{0.20} = 100$, i.e. \$100.

29 **Answer: A** 6.1

Step 1: Distances from 6.06: to 6.1 is 0.04; to 6 is 0.06; to 6.7 is 0.64; to 7 is 0.94.
Step 2: The smallest distance is to 6.1.

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30 **Answer: D** $\frac{60}{1000}$

Step 1: Compare each with $\frac{1}{2}$: $\frac{101}{200} > \frac{100}{200}$, $\frac{17}{33} > \frac{16.5}{33}$, $\frac{7}{12} > \frac{6}{12}$.
Step 2: $\frac{60}{1000} = 0.06 < 0.5$ — the exception.

31 **Answer: D** Greater than 32

Step 1: $\frac{1}{4}x > 8$.
Step 2: Multiply both sides by 4: $x > 32$.

32 **Answer: A** 221 mile/hour

Step 1: Time: 2 h 18 min 30 s $= 2 + \frac{18.5}{60} \approx 2.3083$ hours.
Step 2: Speed $= \frac{510}{2.3083} \approx 220.9 \approx 221$ miles per hour.

33 **Answer: D** 93

Step 1: Numbers of the form $4N + 1$ leave remainder 1 when divided by 4.
Step 2: $55 \to 3$, $62 \to 2$, $71 \to 3$, $93 \to 1$.
Step 3: $93 = 4(23) + 1$.

34 **Answer: D** 18

Step 1: $3(-3)^2 + 5(-3) + 6 = 3(9) - 15 + 6$.
Step 2: $= 27 - 15 + 6 = 18$.

35 **Answer: B** $(2x + 9) + 12x = 90$ because the angles are complementary

Step 1: The figure shows a right angle at B (between $\overrightarrow{BA}$ and $\overrightarrow{BC}$) split into $12x°$ and $(2x + 9)°$.
Step 2: Two angles that together make a right angle are complementary: $(2x + 9) + 12x = 90$.

36 **Answer: D** 14,000

Step 1: 1.75 millionaires per 1000 people.
Step 2: 8,000,000 ÷ 1000 = 8000 groups of 1000.
Step 3: $8000 \times 1.75 = 14{,}000$ millionaires.

37 **Answer: A** 720°

Step 1: Interior angle sum $= (n - 2) \times 180°$.
Step 2: Hexagon: $n = 6 \Rightarrow 4 \times 180° = 720°$.

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38 **Answer: C** 1

Step 1: $x + y = 1 \Rightarrow y = -x + 1$: slope $= -1$, y-intercept $= 1$.
Step 2: x-intercept (set $y = 0$): $x = 1$.
Step 3: Sum $= -1 + 1 + 1 = 1$.

39 **Answer: D** 35

Step 1: Angles of incidence and reflection at the mirror are equal, so $\triangle ABM \sim \triangle CDM$ (both have right angles).
Step 2: $\frac{AB}{BM} = \frac{CD}{DM} \Rightarrow \frac{5}{4} = \frac{h}{28}$.
Step 3: $h = \frac{5 \times 28}{4} = 35$ ft.

40 **Answer: A** The triangles are similar but not congruent

Step 1: Both triangles have angles $55°$, $60°$, $65°$, so they are similar (AA).
Step 2: The 4 m side lies opposite different angles in the two triangles, so corresponding sides are not equal — they are not congruent.

41 **Answer: D** 12

Step 1: The net has 8 equilateral triangles: a regular octahedron.
Step 2: Edges $= \frac{8 \text{ faces} \times 3 \text{ sides}}{2} = 12$ (each edge is shared by two faces).

42 **Answer: A** $\frac{9}{16}$

Step 1: Total cards $= 9 + 7 = 16$.
Step 2: $P(\text{white}) = \frac{9}{16}$.

43 **Answer: B** 11π

Step 1: $C = \pi d$.
Step 2: $C = 11\pi$ inches.

44 **Answer: A** Circle

Step 1: The cutting plane is parallel to the cone's circular base.
Step 2: A cross section parallel to the base of a cone is a circle.

45 **Answer: A** $16x^9y^{15}z^3$

Step 1: $(2x^3y^5z)^3 = 2^3x^9y^{15}z^3 = 8x^9y^{15}z^3$.
Step 2: Multiply by 2: $16x^9y^{15}z^3$.

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46 Answer: **B** $-\frac{2}{9}p^5q^{13}$

Step 1: $\left(\frac{1}{3}p^2q^5\right)^2 = \frac{1}{9}p^4q^{10}$.
Step 2: $-2pq^3 \cdot \frac{1}{9}p^4q^{10} = -\frac{2}{9}p^5q^{13}$.

47 Answer: **A** 227

Step 1: $r = \frac{17}{2} = 8.5$.
Step 2: $A = \pi r^2 = 72.25\pi \approx 226.98 \approx 227$.

48 Answer: **B** $(5x-11)(5x+11)$

Step 1: $25x^2 - 121 = (5x)^2 - 11^2$ is a difference of squares.
Step 2: $a^2 - b^2 = (a-b)(a+b) \Rightarrow (5x-11)(5x+11)$.

49 Answer: **C** 4

Step 1: Pythagorean theorem with legs $\sqrt{x}$ and $2\sqrt{3}$, hypotenuse x: $x + 12 = x^2$.
Step 2: $x^2 - x - 12 = 0 \Rightarrow (x-4)(x+3) = 0$.
Step 3: Length is positive: $x = 4$.

50 Answer: **B** $\frac{3}{2}$

Step 1: The two regular hexagons are similar, with apothems $9\sqrt{3}$ and $6\sqrt{3}$.
Step 2: Perimeters are in the same ratio as any corresponding lengths: $\frac{9\sqrt{3}}{6\sqrt{3}} = \frac{3}{2}$.

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